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DECISION TREES

QUESTIONS

What type of ML task is performed by Decision Trees?

Explain the working procedure of a DT classifier with an example.

Define a decision tree classifier.

What are the components of a decision tree and What are decision rules?

What is the principle/technique behind the functionality of a decision tree classifier explain the technique mathematically.

Define a split point.

Define Data Partition

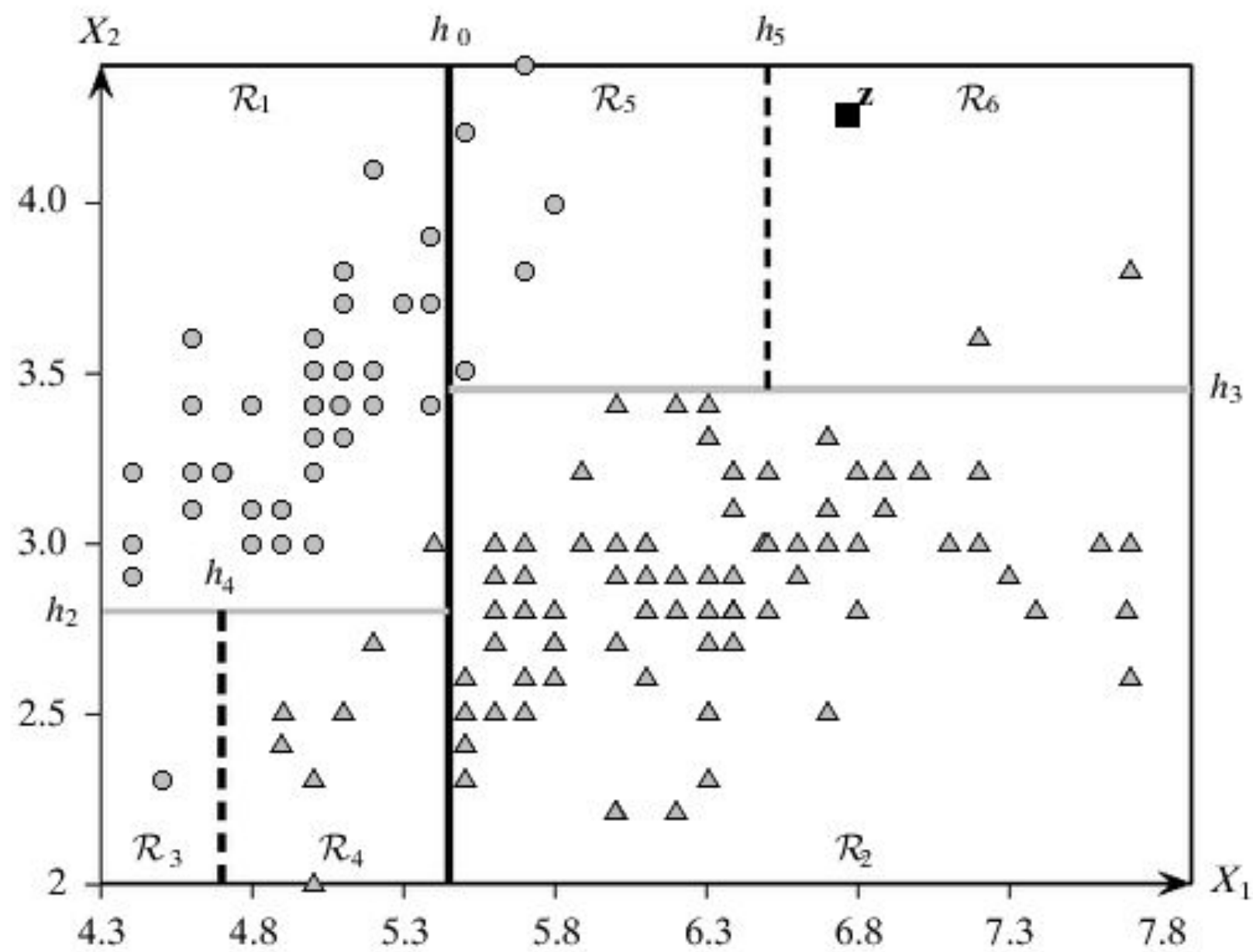
Define purity in the context of a region/space.

How to find the split point in case of a categorical attribute?

What is the stopping condition for splitting?

Problem:

Example 19.1. Consider the Iris dataset shown in Figure 19.1(a), which plots the attributes sepal length (X_1) and sepal width (X_2). The classification task is to discriminate between c_1 , corresponding to iris-setosa (in circles), and c_2 , corresponding to the other two types of Irises (in triangles). The input dataset $\mathbf{D}$ has $n = 150$ points that lie in the data space which is given as the rectangle, $\mathcal{R} = \text{range}(X_1) \times \text{range}(X_2) = [4.3, 7.9] \times [2.0, 4.4]$.



(a) Recursive Splits

A decision tree consists of internal nodes that represent the decisions corresponding to the hyperplanes or split points (i.e., which half-space a given point lies in), and leaf nodes that represent regions or partitions of the data space, which are labeled with the majority class.

A region is characterized by the subset of data points that lie in that region.

$\mathcal{R}_3$: If $X_1 \leq 5.45$ and $X_2 \leq 2.8$ and $X_1 \leq 4.7$, then class is c_1 , or

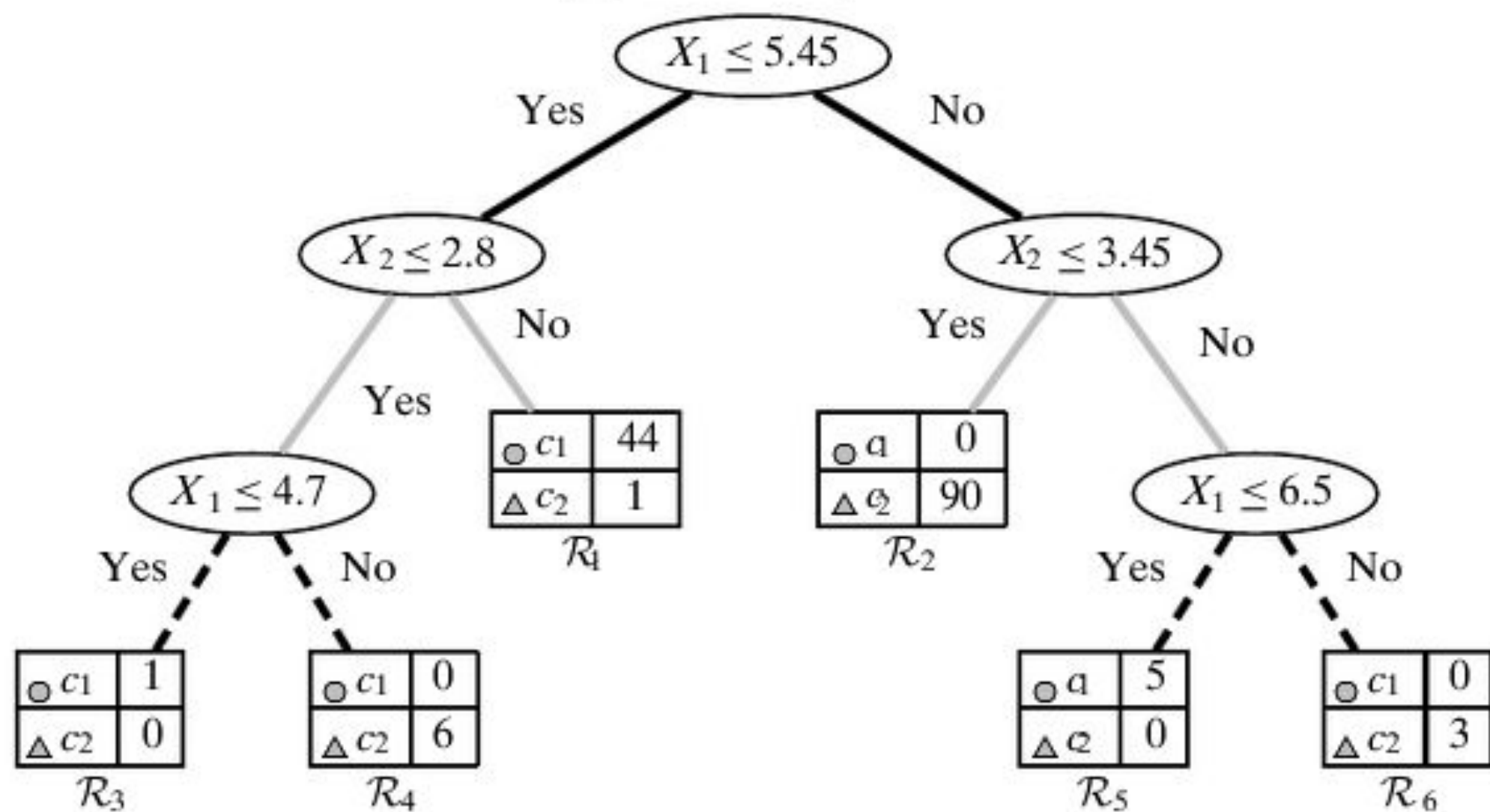
$\mathcal{R}_4$: If $X_1 \leq 5.45$ and $X_2 \leq 2.8$ and $X_1 > 4.7$, then class is c_2 , or

$\mathcal{R}_1$: If $X_1 \leq 5.45$ and $X_2 > 2.8$, then class is c_1 , or

$\mathcal{R}_2$: If $X_1 > 5.45$ and $X_2 \leq 3.45$, then class is c_2 , or

$\mathcal{R}_5$: If $X_1 > 5.45$ and $X_2 > 3.45$ and $X_1 \leq 6.5$, then class is c_1 , or

$\mathcal{R}_6$: If $X_1 > 5.45$ and $X_2 > 3.45$ and $X_1 > 6.5$, then class is c_2



(b) Decision Tree

Axis-Parallel Hyperplanes

A hyperplane $h(\mathbf{x})$ is defined as the set of all points $\mathbf{x}$ that satisfy the following equation

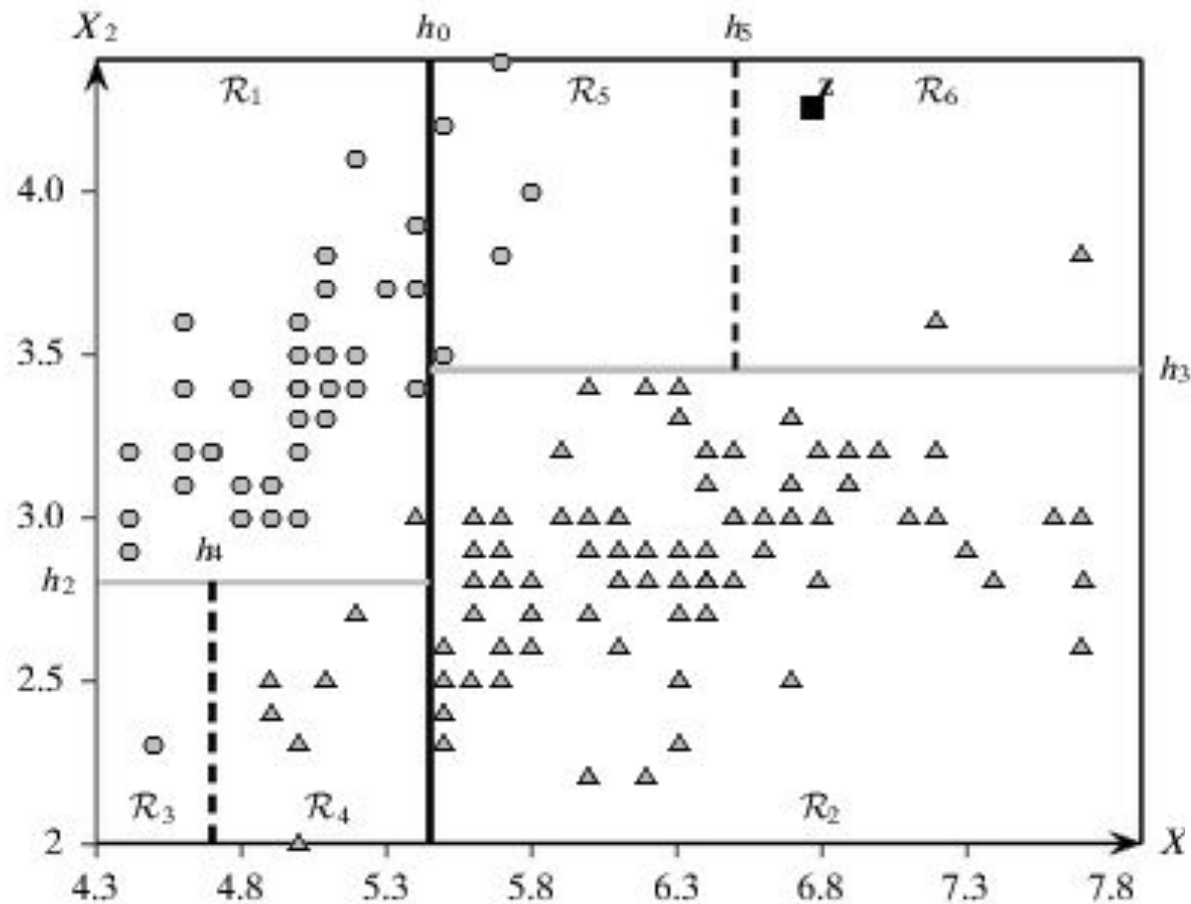
$$h(\mathbf{x}): \mathbf{w}^T \mathbf{x} + b = 0 \quad (19.1)$$

Here $\mathbf{w} \in \mathbb{R}^d$ is a *weight vector* that is normal to the hyperplane, and b is the offset of the hyperplane from the origin. A decision tree considers only *axis-parallel hyperplanes*, that is, the weight vector must be parallel to one of the original dimensions or axes X_j . Put differently, the weight vector $\mathbf{w}$ is restricted *a priori* to one of the standard basis vectors $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_d\}$, where $\mathbf{e}_j \in \mathbb{R}^d$ has a 1 for the j th dimension, and 0 for all other dimensions. If $\mathbf{x} = (x_1, x_2, \dots, x_d)^T$ and assuming $\mathbf{w} = \mathbf{e}_j$, we can rewrite Eq. (19.1) as

$$h(\mathbf{x}): \mathbf{e}_j^T \mathbf{x} + b = 0, \text{ which implies that}$$

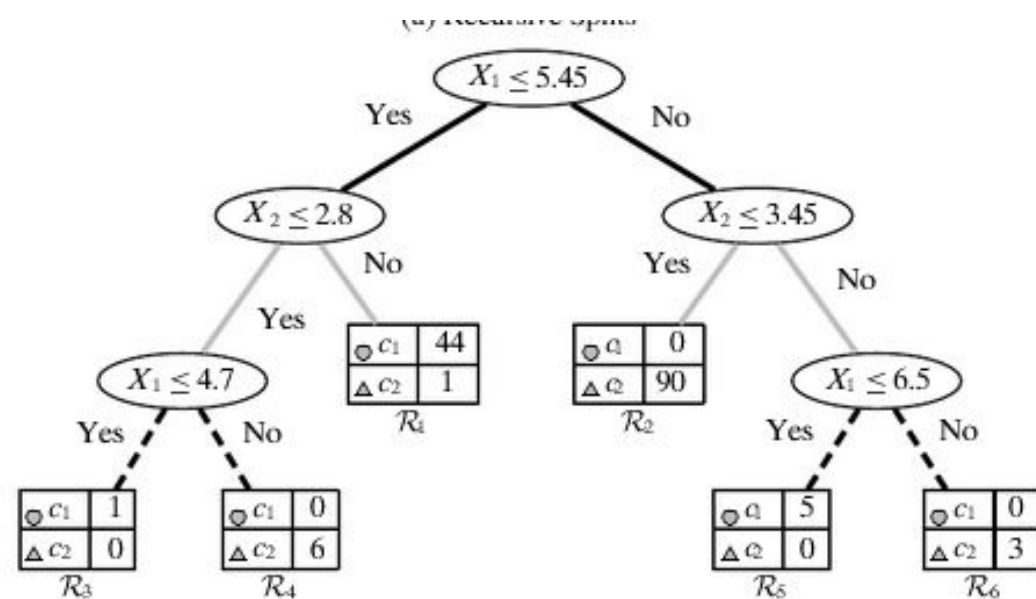
$$h(\mathbf{x}): x_j + b = 0$$

where the choice of the offset b yields different hyperplanes along dimension X_j .



(a) Recursive Splits

$\mathcal{R}_3$: If $X_1 \leq 5.45$ and $X_2 \leq 2.8$ and $X_1 \leq 4.7$, then class is c_1 , or
 $\mathcal{R}_4$: If $X_1 \leq 5.45$ and $X_2 \leq 2.8$ and $X_1 > 4.7$, then class is c_2 , or
 $\mathcal{R}_1$: If $X_1 \leq 5.45$ and $X_2 > 2.8$, then class is c_1 , or
 $\mathcal{R}_2$: If $X_1 > 5.45$ and $X_2 \leq 3.45$, then class is c_2 , or
 $\mathcal{R}_5$: If $X_1 > 5.45$ and $X_2 > 3.45$ and $X_1 \leq 6.5$, then class is c_1 , or
 $\mathcal{R}_6$: If $X_1 > 5.45$ and $X_2 > 3.45$ and $X_1 > 6.5$, then class is c_2



(b) Decision Tree

Figure 19.1. Decision trees: recursive partitioning via axis-parallel hyperplanes.

Keywords

- Split Point
- Data Partition
- Purity

Split Point for Numeric Attributes

$$X_j \leq v$$

where $v = -b$ is some value in the domain of attribute X_j . The decision or split point $X_j \leq v$ thus splits the input data space $\mathcal{R}$ into two regions $\mathcal{R}_Y$ and $\mathcal{R}_N$, which denote the set of *all possible points* that satisfy the decision and those that do not.

Data Partition

Each split of $\mathcal{R}$ into $\mathcal{R}_Y$ and $\mathcal{R}_N$ also induces a binary partition of the corresponding input data points $\mathbf{D}$. That is, a split point of the form $X_j \leq v$ induces the data partition

$$\mathbf{D}_Y = \{\mathbf{x}^T \mid \mathbf{x} \in \mathbf{D}, x_j \leq v\}$$

$$\mathbf{D}_N = \{\mathbf{x}^T \mid \mathbf{x} \in \mathbf{D}, x_j > v\}$$

Purity

$$\textit{purity}(\mathbf{D}_j) = \max_i \left\{ \frac{n_{ji}}{n_j} \right\}$$

where $n_j = |\mathbf{D}_j|$ is the total number of data points in the region $\mathcal{R}_j$, and n_{ji} is the number of points in $\mathbf{D}_j$ with class label c_i .

Categorical Variables

Categorical Attributes

In addition to numeric attributes, a decision tree can also handle categorical data. For a categorical attribute X_j , the split points or decisions are of the $X_j \in V$, where $V \subset \text{dom}(X_j)$, and $\text{dom}(X_j)$ denotes the domain for X_j . Intuitively, this split can be considered to be the categorical analog of a hyperplane. It results in two “half-spaces,” one region $\mathcal{R}_Y$ consisting of points $\mathbf{x}$ that satisfy the condition $x_i \in V$, and the other region $\mathcal{R}_N$ comprising points that satisfy the condition $x_i \notin V$.

For a **categorical attribute** x_j (e.g., color, size, type), the split is of the form:

$$X_j \in V$$

where v is a **subset** of the domain $\text{dom}(x_j)$. This is equivalent to splitting the data into:

- **RY**: data points where the attribute value is in v
- **RN**: data points where the attribute value is **not** in v

This is similar to a hyperplane for numeric data but applied to **discrete sets**.

TRAINING EXAMPLE:

Day	Outlook	Temp	Humidity	Wind	Tennis?
<i>D1</i>	Sunny	Hot	High	Weak	<i>No</i>
<i>D2</i>	Sunny	Hot	High	Strong	<i>No</i>
<i>D3</i>	Overcast	Hot	High	Weak	<i>Yes</i>
<i>D4</i>	Rain	Mild	High	Weak	<i>Yes</i>
<i>D5</i>	Rain	Cool	Normal	Weak	<i>Yes</i>
<i>D6</i>	Rain	Cool	Normal	Strong	<i>No</i>
<i>D7</i>	Overcast	Cool	Normal	Strong	<i>Yes</i>
<i>D8</i>	Sunny	Mild	High	Weak	<i>No</i>
<i>D9</i>	Sunny	Cool	Normal	Weak	<i>Yes</i>
<i>D10</i>	Rain	Mild	Normal	Weak	<i>Yes</i>
<i>D11</i>	Sunny	Mild	Normal	Strong	<i>Yes</i>
<i>D12</i>	Overcast	Mild	High	Strong	<i>Yes</i>
<i>D13</i>	Overcast	Hot	Normal	Weak	<i>Yes</i>
<i>D14</i>	Rain	Mild	High	Strong	<i>No</i>

Day	Outlook	Temp	Humidity	Wind	Tennis?
<i>D1</i>	Sunny	Hot	High	Weak	<i>No</i>
<i>D2</i>	Sunny	Hot	High	Strong	<i>No</i>
<i>D3</i>	Overcast	Hot	High	Weak	<i>Yes</i>
<i>D4</i>	Rain	Mild	High	Weak	<i>Yes</i>
<i>D5</i>	Rain	Cool	Normal	Weak	<i>Yes</i>
<i>D6</i>	Rain	Cool	Normal	Strong	<i>No</i>
<i>D7</i>	Overcast	Cool	Normal	Strong	<i>Yes</i>
<i>D8</i>	Sunny	Mild	High	Weak	<i>No</i>
<i>D9</i>	Sunny	Cool	Normal	Weak	<i>Yes</i>
<i>D10</i>	Rain	Mild	Normal	Weak	<i>Yes</i>
<i>D11</i>	Sunny	Mild	Normal	Strong	<i>Yes</i>
<i>D12</i>	Overcast	Mild	High	Strong	<i>Yes</i>
<i>D13</i>	Overcast	Hot	Normal	Weak	<i>Yes</i>
<i>D14</i>	Rain	Mild	High	Strong	<i>No</i>

The target class: **PlayTennis**
(Yes/No)

We will focus on the categorical
attribute: **Outlook**

Day	Outlook	Temp	Humidity	Wind	Tennis?
<i>D1</i>	Sunny	Hot	High	Weak	<i>No</i>
<i>D2</i>	Sunny	Hot	High	Strong	<i>No</i>
<i>D3</i>	Overcast	Hot	High	Weak	<i>Yes</i>
<i>D4</i>	Rain	Mild	High	Weak	<i>Yes</i>
<i>D5</i>	Rain	Cool	Normal	Weak	<i>Yes</i>
<i>D6</i>	Rain	Cool	Normal	Strong	<i>No</i>
<i>D7</i>	Overcast	Cool	Normal	Strong	<i>Yes</i>
<i>D8</i>	Sunny	Mild	High	Weak	<i>No</i>
<i>D9</i>	Sunny	Cool	Normal	Weak	<i>Yes</i>
<i>D10</i>	Rain	Mild	Normal	Weak	<i>Yes</i>
<i>D11</i>	Sunny	Mild	Normal	Strong	<i>Yes</i>
<i>D12</i>	Overcast	Mild	High	Strong	<i>Yes</i>
<i>D13</i>	Overcast	Hot	Normal	Weak	<i>Yes</i>
<i>D14</i>	Rain	Mild	High	Strong	<i>No</i>

For the attribute **Outlook**, the domain is:

$$\text{dom}(X_j) = \{\text{Sunny}, \text{Overcast}, \text{Rain}\}$$

To split this categorical attribute, we consider binary splits such as:

- $X_j \in \{\text{Sunny}\}$
- $X_j \in \{\text{Overcast}\}$
- $X_j \in \{\text{Rain}\}$
- $X_j \in \{\text{Sunny}, \text{Overcast}\}$
- $X_j \in \{\text{Sunny}, \text{Rain}\}$
- $X_j \in \{\text{Overcast}, \text{Rain}\}$

Suppose we evaluate the split:

$$\text{Outlook} \in \{\text{Sunny}\}$$

Then:

- Region **RY** = rows where Outlook = Sunny (5 rows)
- Region **RN** = rows where Outlook $\neq$ Sunny (9 rows)

DATA SET:

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

DECISION_TREE(D, η, π):

1. $n \leftarrow |D|$
// Total number of data points in the dataset D
2. For each class c_i :
3. $n_i \leftarrow |\{x \in D \mid y = c_i\}|$
// Count how many data points belong to class c_i
4. $\text{purity}(D) \leftarrow \max_i(n_i / n)$
// Compute the purity: proportion of the dominant class
5. If $n \leq \eta$ or $\text{purity}(D) \geq \pi$ then
// Stopping condition: stop if dataset is too small or already pure
6. $c^* \leftarrow \text{argmax}_{c_i}(n_i / n)$
// Determine the majority class in the current partition
7. Create a leaf node and label it with class c^*
// Stop splitting and assign the majority class label to the leaf
8. Return
// End recursion for this node

9. $(\text{split_point}^*, \text{score}^*) \leftarrow (\emptyset, 0)$
// Initialize the best split point and its score
10. For each attribute $X_{\square}$ in D :
// Consider each attribute (feature) to find the best split
11. If $X_{\square}$ is numeric:
// Check if the attribute is numerical
12. $(v, \text{score}) \leftarrow \text{EVALUATE-NUMERIC-ATTRIBUTE}(D, X_{\square})$
// Evaluate the best threshold value `v` for splitting $X_{\square}$
13. If $\text{score} > \text{score}^*$:
// If the resulting score is better than the current best
14. $(\text{split_point}^*, \text{score}^*) \leftarrow (X_{\square} \leq v, \text{score})$
// Update the best split point to $X_{\square} \leq v$

15. Else if $X_{\square}$ is categorical:
// If the attribute is categorical
16. $(V, \text{score}) \leftarrow \text{EVALUATE-CATEGORICAL-ATTRIBUTE}(D, X_{\square})$
// Evaluate the best subset V of categories for splitting
17. If $\text{score} > \text{score}^*$:
// Check if this split is better than the current best
18. $(\text{split_point}^*, \text{score}^*) \leftarrow (X_{\square} \in V, \text{score})$
// Update the best split point to $X_{\square} \in V$
19. $D_Y \leftarrow \{x \in D \mid x \text{ satisfies } \text{split_point}^*\}$
// Create left subset: data points that match the best split condition
20. $D_N \leftarrow \{x \in D \mid x \text{ does not satisfy } \text{split_point}^*\}$
// Create right subset: data points that don't match the split
21. Create an internal node for split_point^* , with two children:
 - Left: $\text{DECISIONTREE}(D_Y, \eta, \pi)$ // Recursively split left subset
 - Right: $\text{DECISIONTREE}(D_N, \eta, \pi)$ // Recursively split right subset

What is this Score:

- Entropy
- Gini Index
- Information Gain

Entropy

- Entropy, in general, measures the amount of disorder or uncertainty in a system.
- A partition/Region has lower entropy (or low disorder) → if Relatively pure → Most of the points have the same label.
- A partition/Region has higher entropy (or more disorder) → if the class labels are mixed → and there is no majority class as such

Formula

The entropy of a set of labeled points $\mathbf{D}$ is defined as follows:

$$H(\mathbf{D}) = - \sum_{i=1}^k P(c_i|\mathbf{D}) \log_2 P(c_i|\mathbf{D})$$

DATA SET for Entropy based DT:

Day	Outlook	Temp	Humidity	Wind	Tennis?
<i>D1</i>	Sunny	Hot	High	Weak	<i>No</i>
<i>D2</i>	Sunny	Hot	High	Strong	<i>No</i>
<i>D3</i>	Overcast	Hot	High	Weak	<i>Yes</i>
<i>D4</i>	Rain	Mild	High	Weak	<i>Yes</i>
<i>D5</i>	Rain	Cool	Normal	Weak	<i>Yes</i>
<i>D6</i>	Rain	Cool	Normal	Strong	<i>No</i>
<i>D7</i>	Overcast	Cool	Normal	Strong	<i>Yes</i>
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<i>D14</i>	Rain	Mild	High	Strong	<i>No</i>

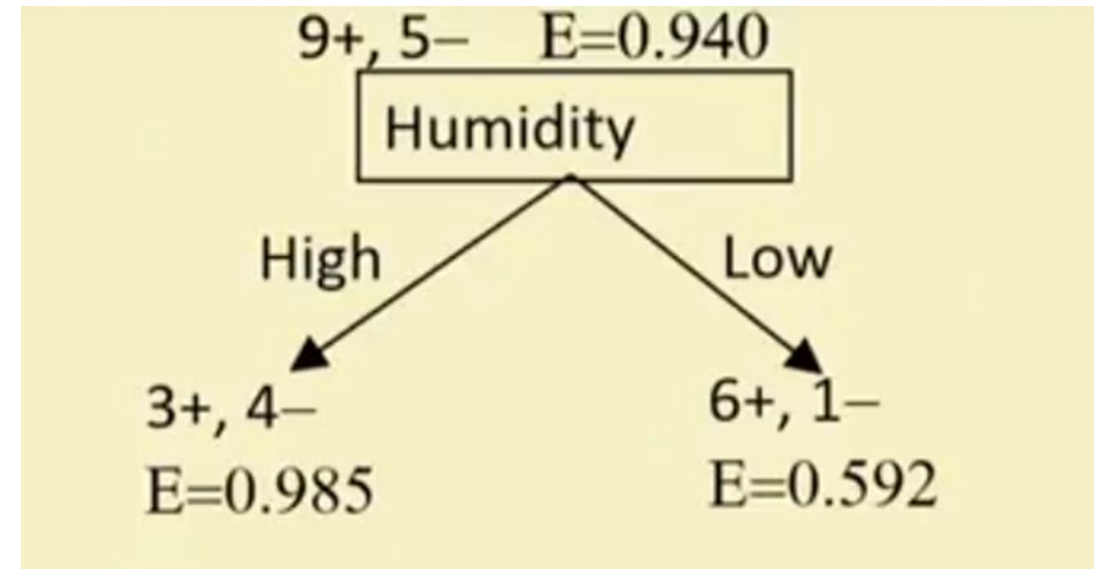
Entropy Based Calculation

$$H(\mathbf{D}) = - \sum_{i=1}^k P(c_i|\mathbf{D}) \log_2 P(c_i|\mathbf{D})$$

Note: $P(c_i|\mathbf{D})$: Denotes the fractions of record belonging to class “i” at a given partition “D”

$$-\frac{9}{14} \log \frac{9}{14} - \frac{5}{14} \log \frac{5}{14} = 0.940$$

Day	Outlook	Temp	Humidity	Wind	Tennis?
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No



$$-\frac{3}{7} \log \frac{3}{7} - \frac{4}{7} \log \frac{4}{7} = 0.985$$

$$-\frac{6}{7} \log \frac{6}{7} - \frac{1}{7} \log \frac{1}{7} = 0.592$$

Some conclusions

- If a region is pure, that is, has points from the same class, then the entropy is zero.
- If the classes are all mixed up, and each appears with equal probability $P(c_i|D) = 1/k$
- Then the entropy has the highest value, $H(D) = \log_2 k$.

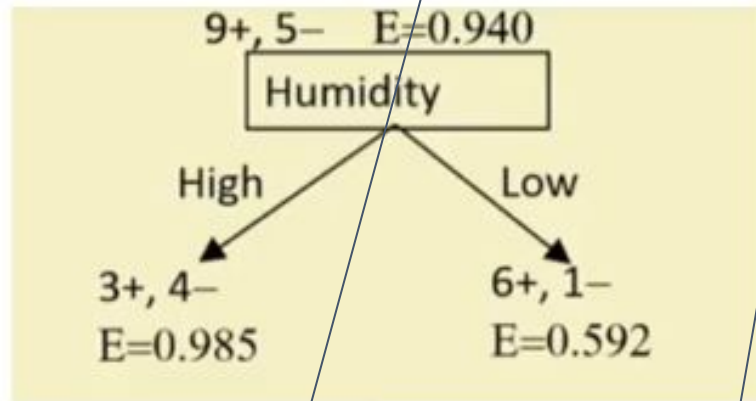
Split Entropy as Weighted Entropy

Higher the information gain $\rightarrow$ the more the reduction in entropy $\rightarrow$ better the split point.

$$H(\mathbf{D}_Y, \mathbf{D}_N) = \frac{n_Y}{n} H(\mathbf{D}_Y) + \frac{n_N}{n} H(\mathbf{D}_N)$$

$$\text{Gain}(\mathbf{D}, \mathbf{D}_Y, \mathbf{D}_N) = H(\mathbf{D}) - H(\mathbf{D}_Y, \mathbf{D}_N)$$

$$-\frac{9}{14} \log \frac{9}{14} - \frac{5}{14} \log \frac{5}{14} = 0.940$$

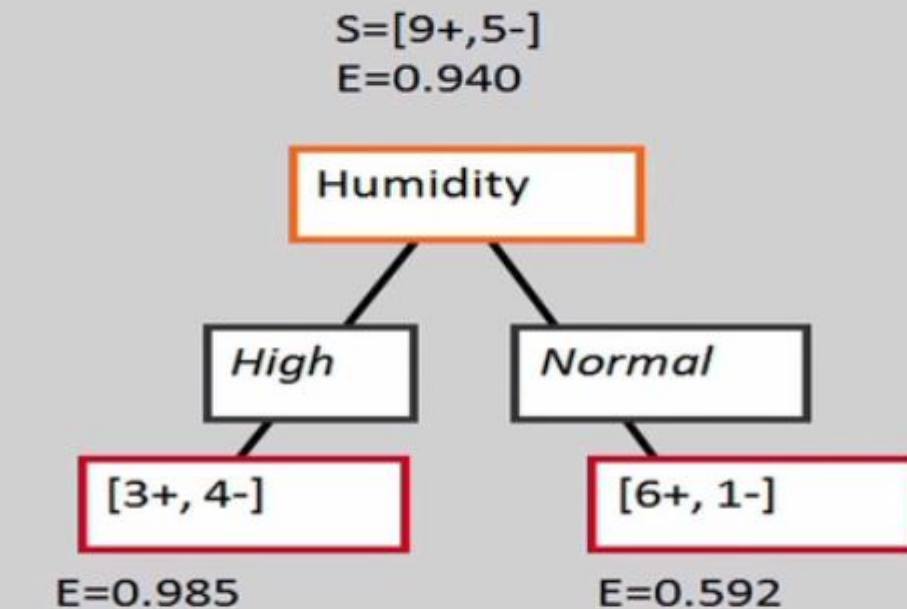


$$-\frac{3}{7} \log \frac{3}{7} - \frac{4}{7} \log \frac{4}{7} = 0.985$$

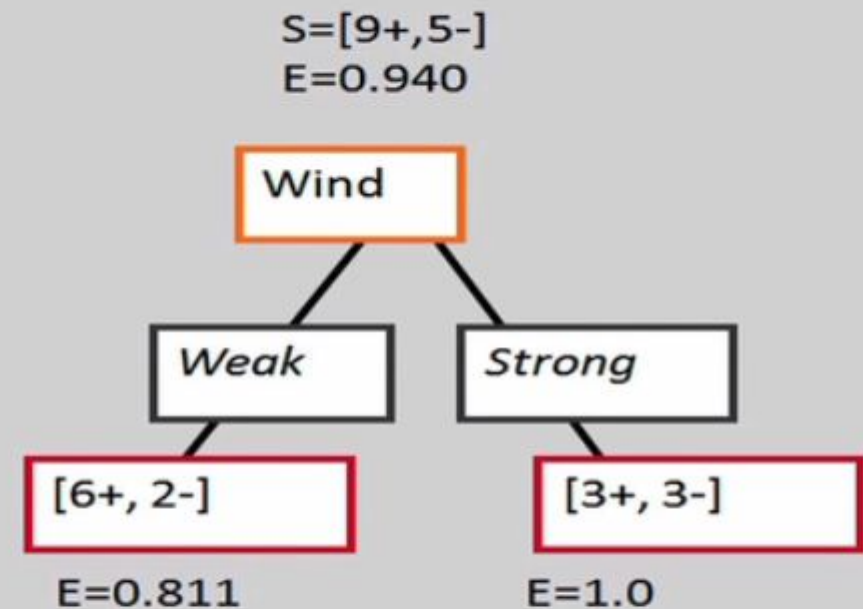
$$-\frac{6}{7} \log \frac{6}{7} - \frac{1}{7} \log \frac{1}{7} = 0.592$$

$$\begin{aligned} \text{Gain}(S, \text{Humidity}) &= 0.940 - (7/14) * 0.985 \\ &\quad - (7/14) * 0.592 \\ &= 0.151 \end{aligned}$$

Selecting the Next Attribute



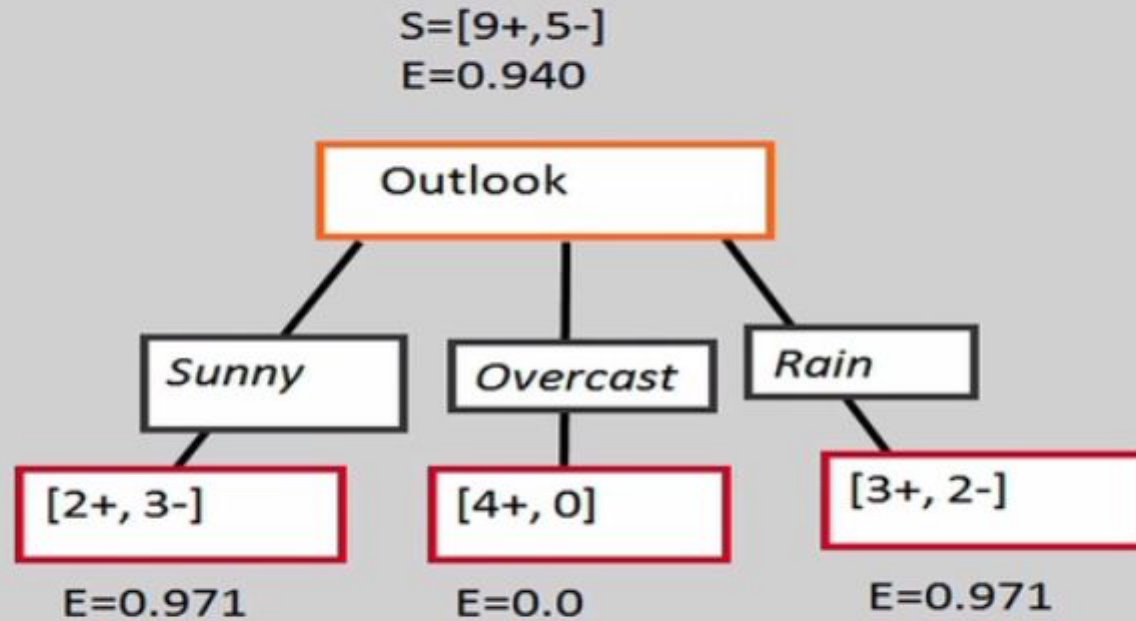
$$\begin{aligned}\text{Gain}(S, \text{Humidity}) &= 0.940 - (7/14) * 0.985 \\ &\quad - (7/14) * 0.592 \\ &= 0.151\end{aligned}$$



$$\begin{aligned}\text{Gain}(S, \text{Wind}) &= 0.940 - (8/14) * 0.811 \\ &\quad - (6/14) * 1.0 \\ &= 0.048\end{aligned}$$

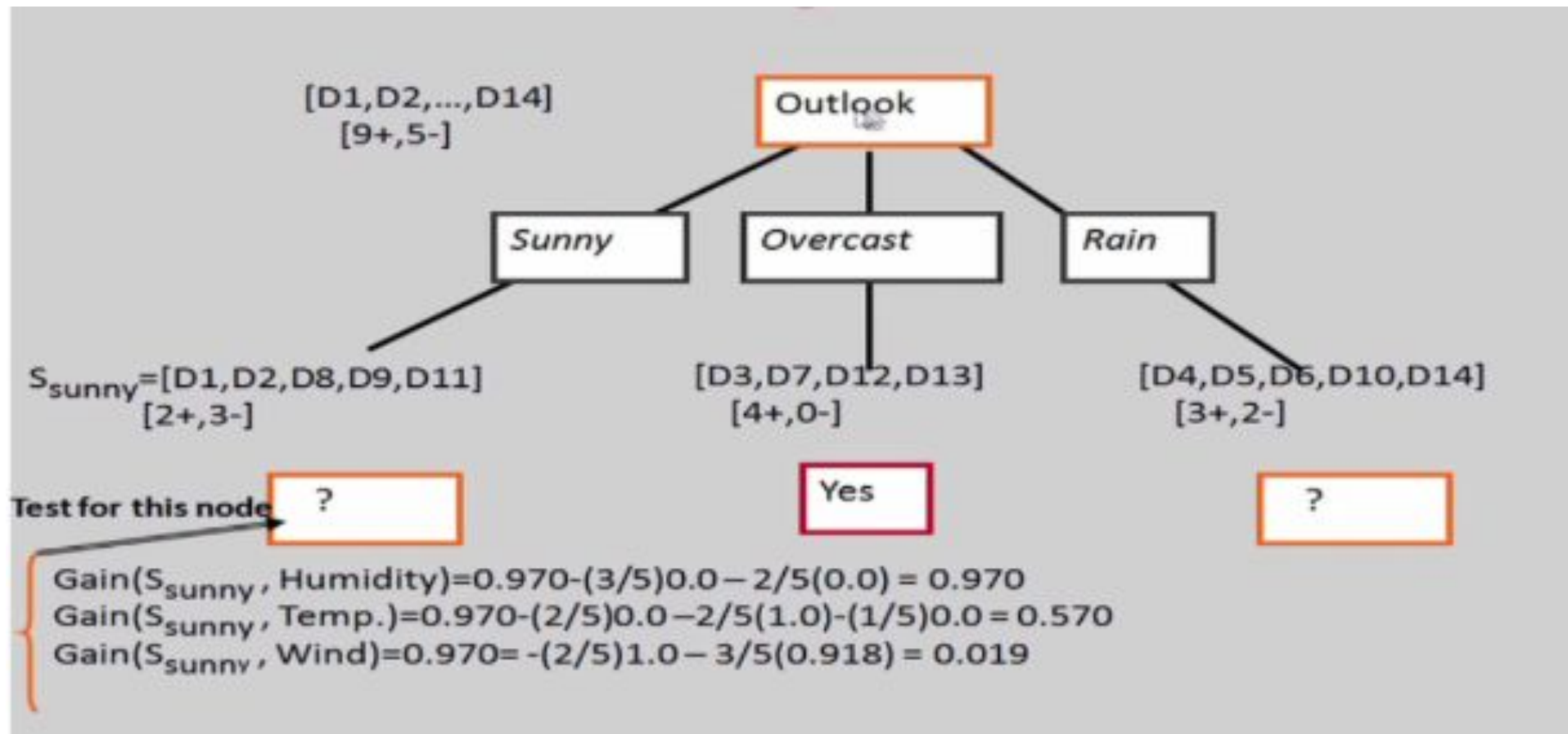
Humidity provides greater info. gain than Wind, w.r.t target classification.

Selecting the Next Attribute

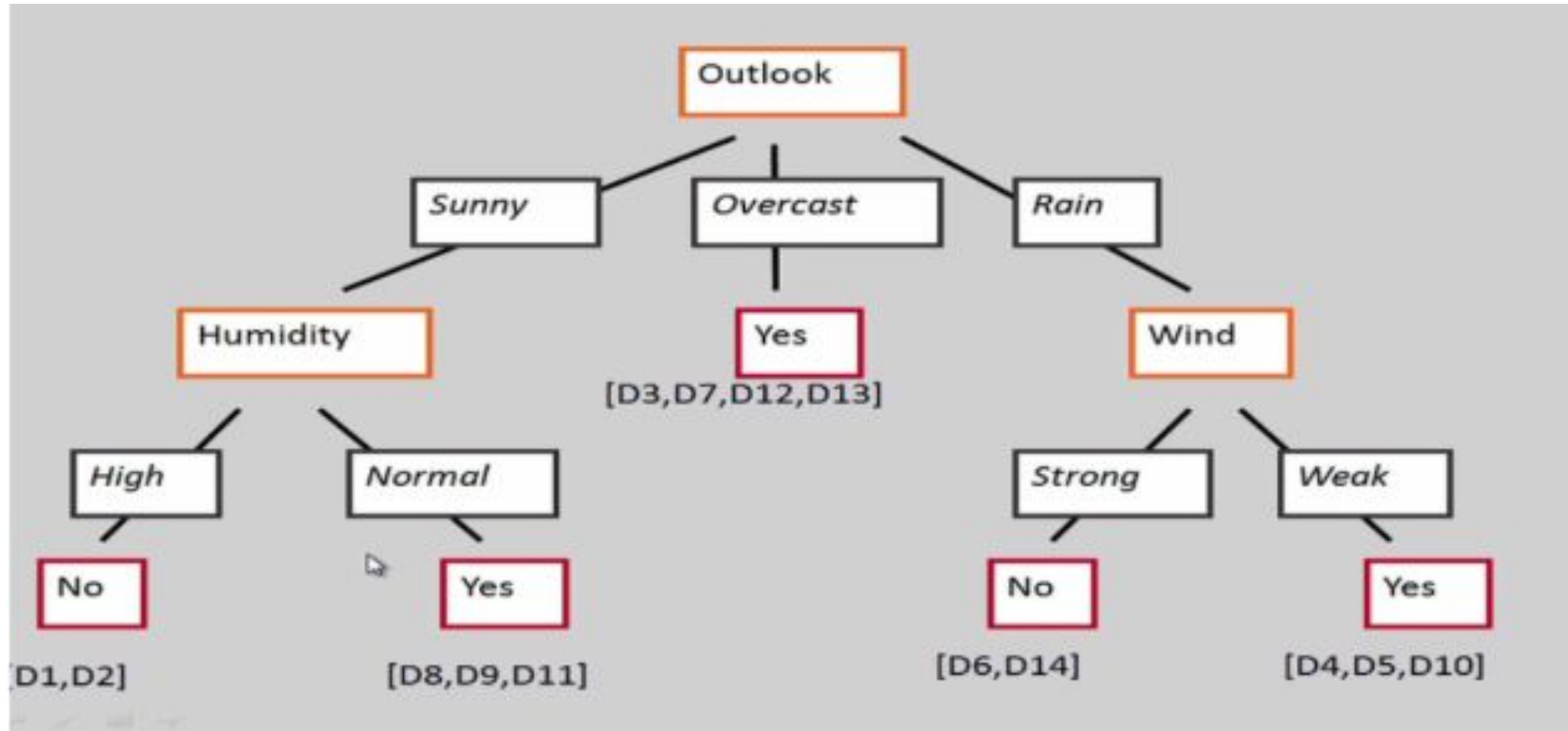


$$\begin{aligned} \text{Gain}(S, \text{Outlook}) &= 0.940 - (5/14) * 0.971 \\ &\quad - (4/14) * 0.0 - (5/14) * 0.971 \\ &= 0.247 \end{aligned}$$

Next Level



Next Level

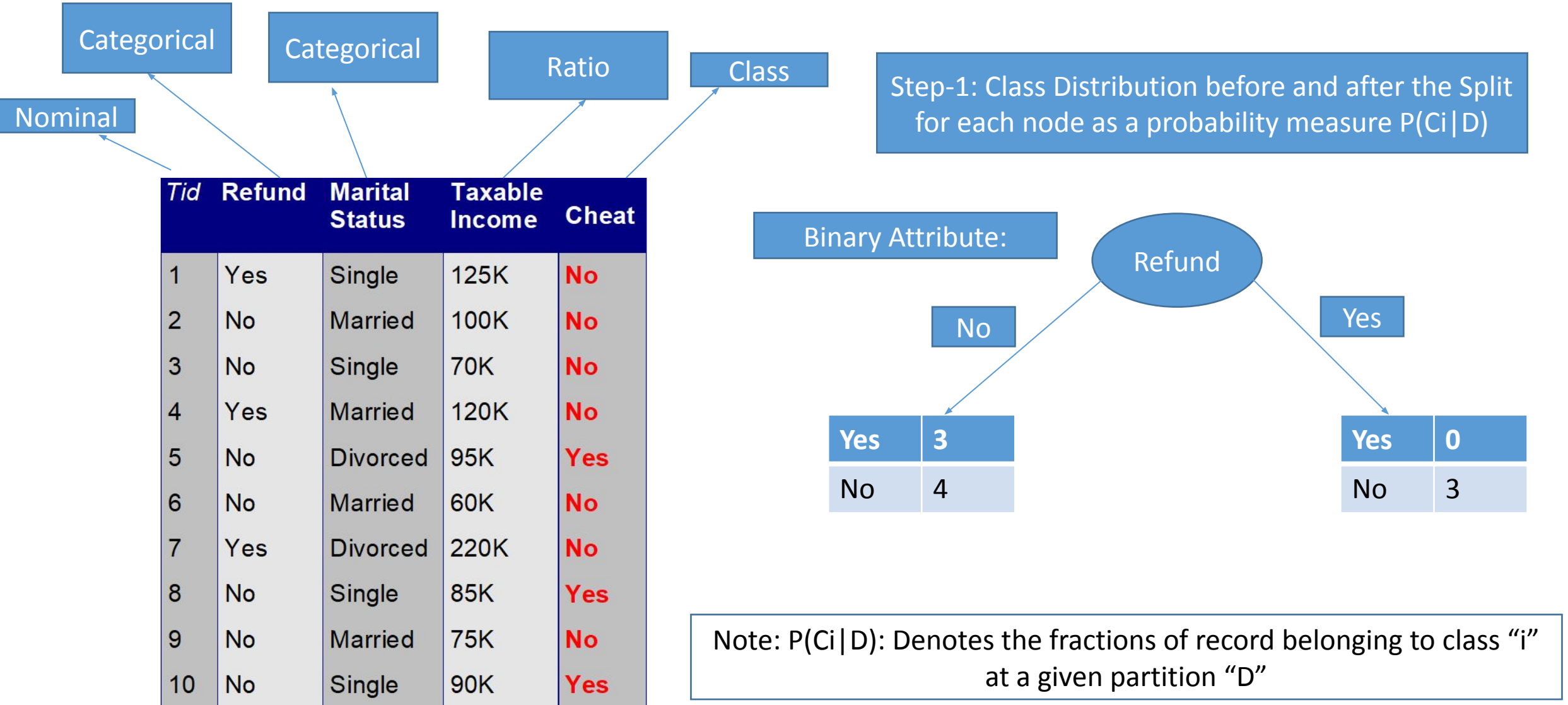


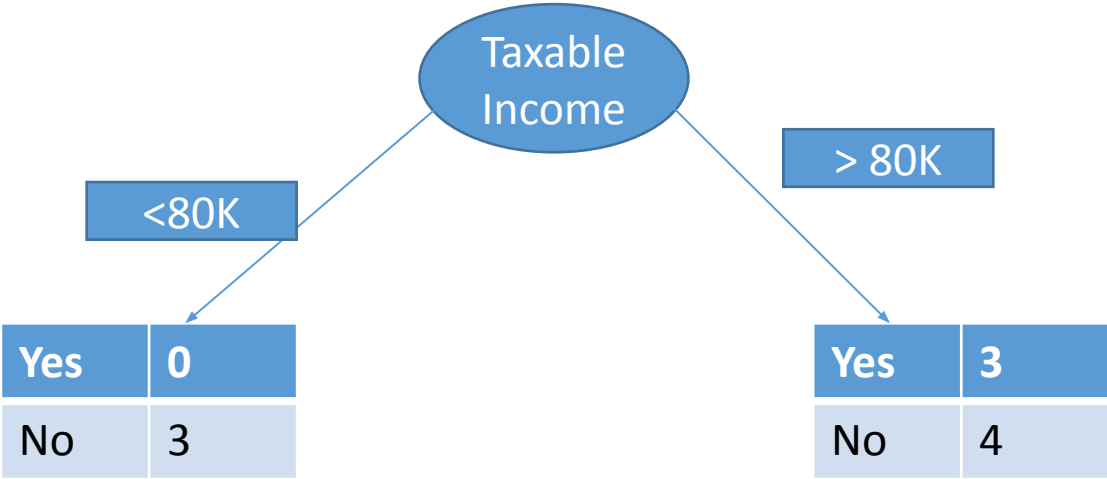
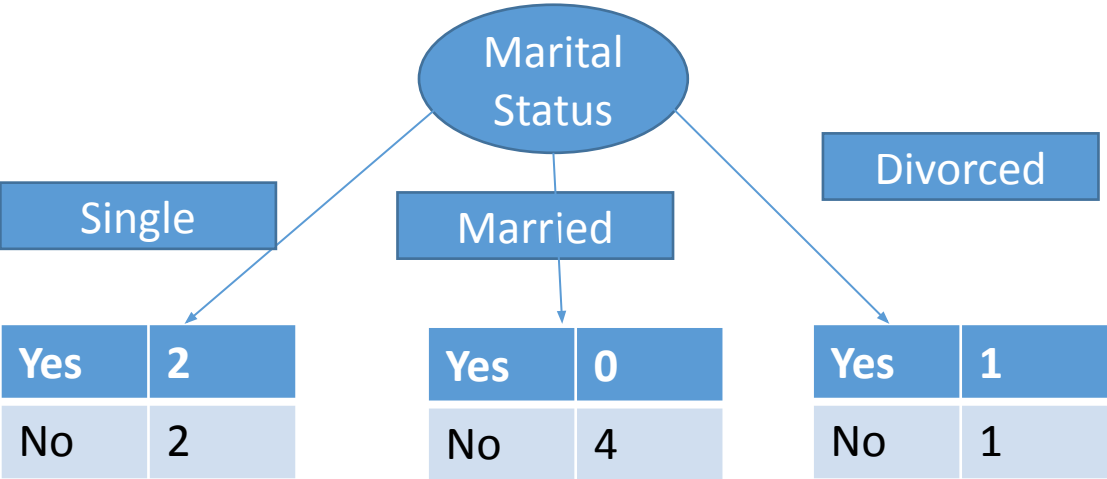
Measuring Gini

$$G(\mathbf{D}) = 1 - \sum_{i=1}^k P(c_i | \mathbf{D})^2$$

- If the partition is pure, then the probability of the majority class is 1 and the probability of all other classes is 0 , and thus, the Gini index is 0.
- On the other hand, when each class is equally represented, with probability $P(c_i|D) = 1/k$
- Then the Gini index has value $k-1/k$
- Thus, higher values of the Gini index indicate more disorder, and lower values indicate more order in terms of the class labels.

Steps for measuring Gini Index:





Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Parent Node(Refund)	
Yes	3
No	7
Gini	

$$1 - \left(\frac{3}{10}\right)^2 - \left(\frac{7}{10}\right)^2 = 0.42$$

Before Splitting at Parent Node

	Dy(Yes)	Dn(No)
Yes	3	0
No	4	3

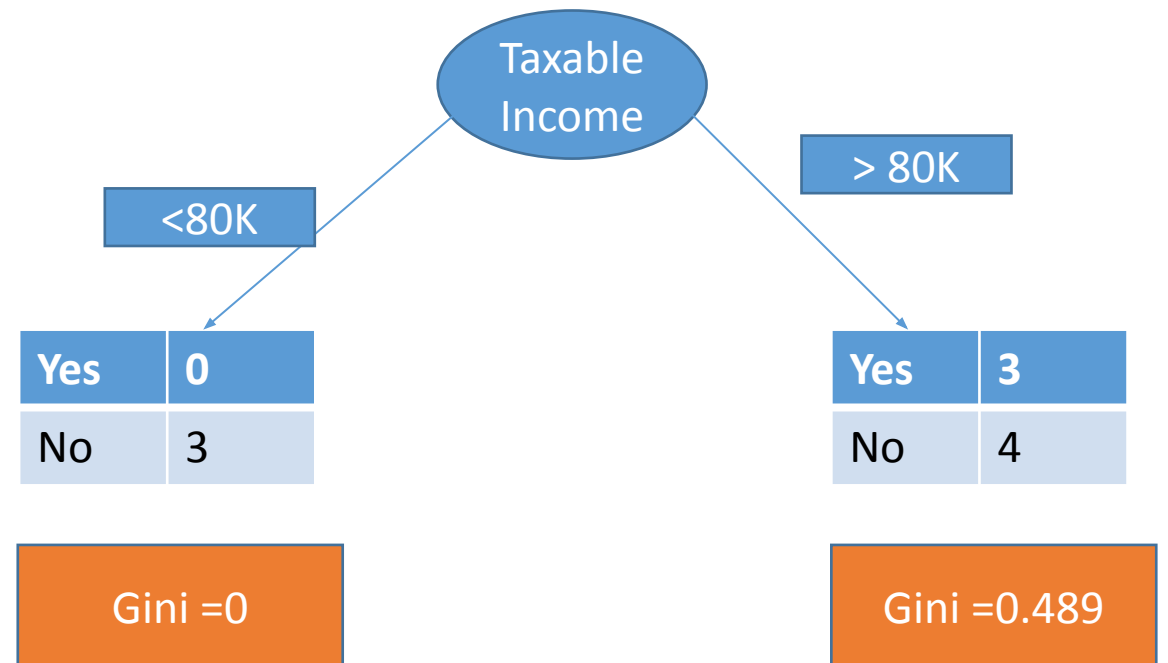
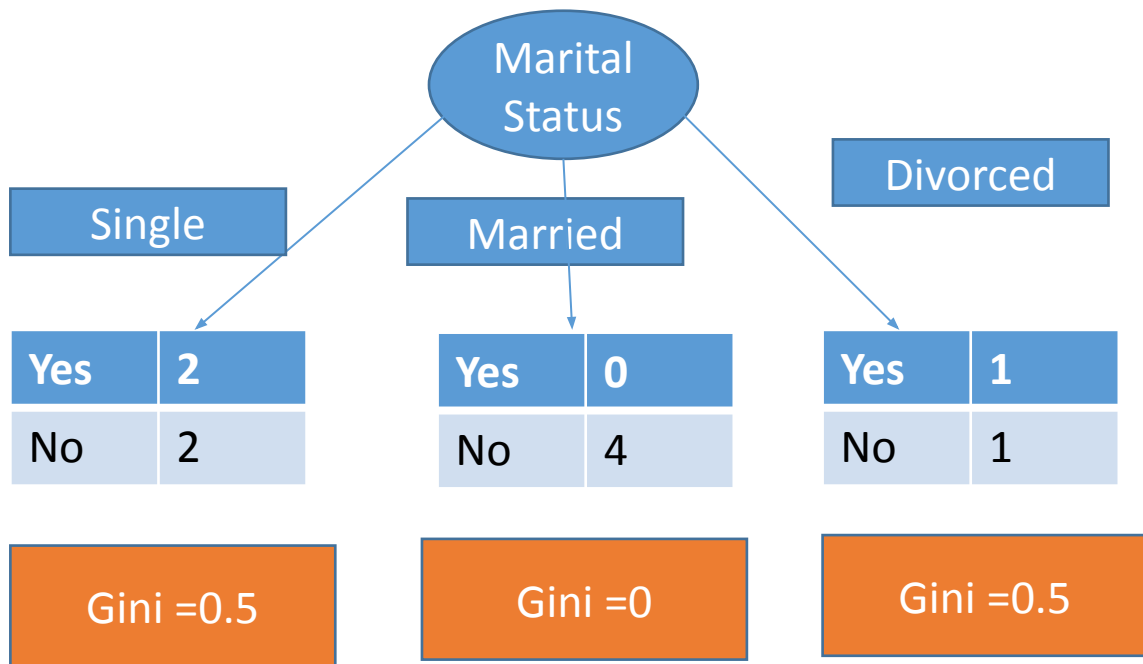
$$1 - \left(\frac{3}{7}\right)^2 - \left(\frac{4}{7}\right)^2 = 0.489$$

For Dy

$$1 - \left(\frac{0}{3}\right)^2 - \left(\frac{3}{3}\right)^2 = 0$$

For Dn

$$G(\mathbf{D}) = 1 - \sum_{i=1}^k P(c_i | \mathbf{D})^2$$



Split Point or Average Weight

We can compute the weighted Gini index of a split point as follows:

$$G(\mathbf{D}_Y, \mathbf{D}_N) = \frac{n_Y}{n}G(\mathbf{D}_Y) + \frac{n_N}{n}G(\mathbf{D}_N)$$

Average Weighted Measure

Refund

No

Yes

Yes	3
No	4

0.489

Yes	0
No	3

0

$$\frac{7}{10} \times 0.489 + \frac{3}{10} \times 0 = 0.34$$

Marital Status

Single,
Divorced

Married

Yes	3
No	1

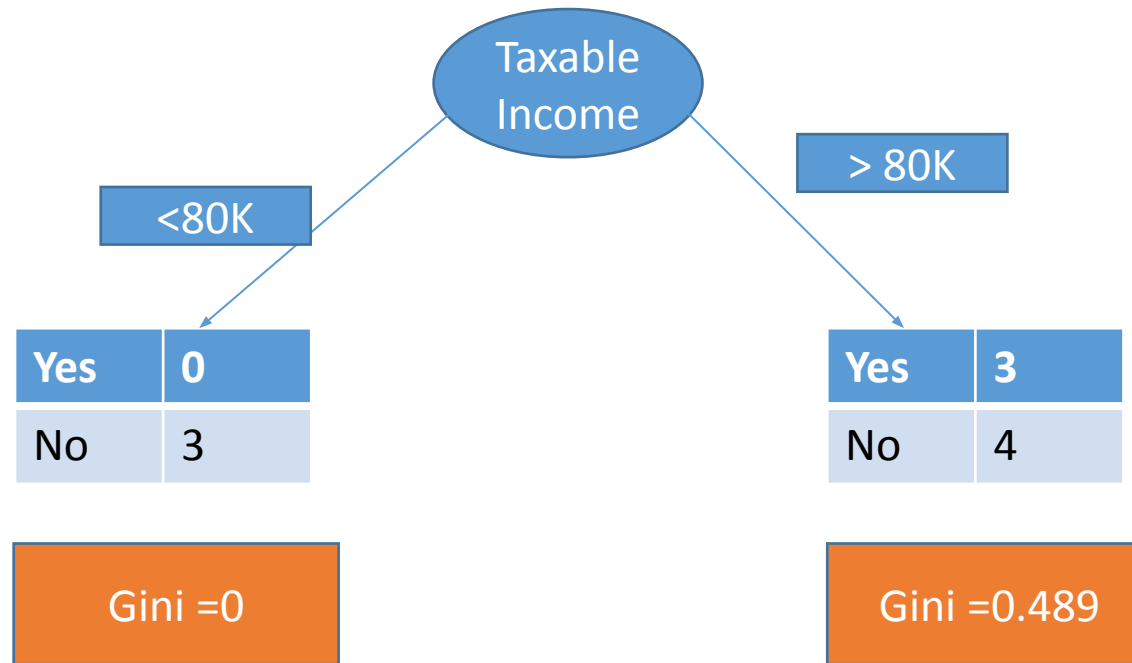
0.37

Yes	0
No	3

0

$$\frac{4}{10} \times 0.37 + \frac{3}{10} \times 0 = 0.138$$

Average Weighted Measure



$$\frac{7}{10} \times 0.489 + \frac{3}{10} \times 0 = 0.34$$

Also CART

$$CART(\mathbf{D}_Y, \mathbf{D}_N) = 2 \frac{n_Y}{n} \frac{n_N}{n} \sum_{i=1}^k \left| P(c_i | \mathbf{D}_Y) - P(c_i | \mathbf{D}_N) \right|$$

Splitting Based on Nominal Attributes

- **Multi-way split:** Use as many partitions as distinct values.



- **Binary split:** Divides values into two subsets. Need to find optimal partitioning.



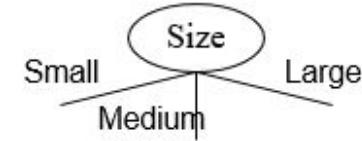
Splitting Continuous attribute in two ways



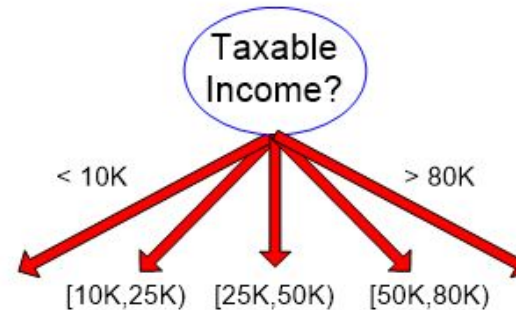
(i) Binary split

Splitting Based on Ordinal Attributes

- **Multi-way split:** Use as many partitions as distinct values.



- **Binary split:** Divides values into two subsets. Need to find optimal partitioning.



(ii) Multi-way split